

Implementation

Creating the Scenario.

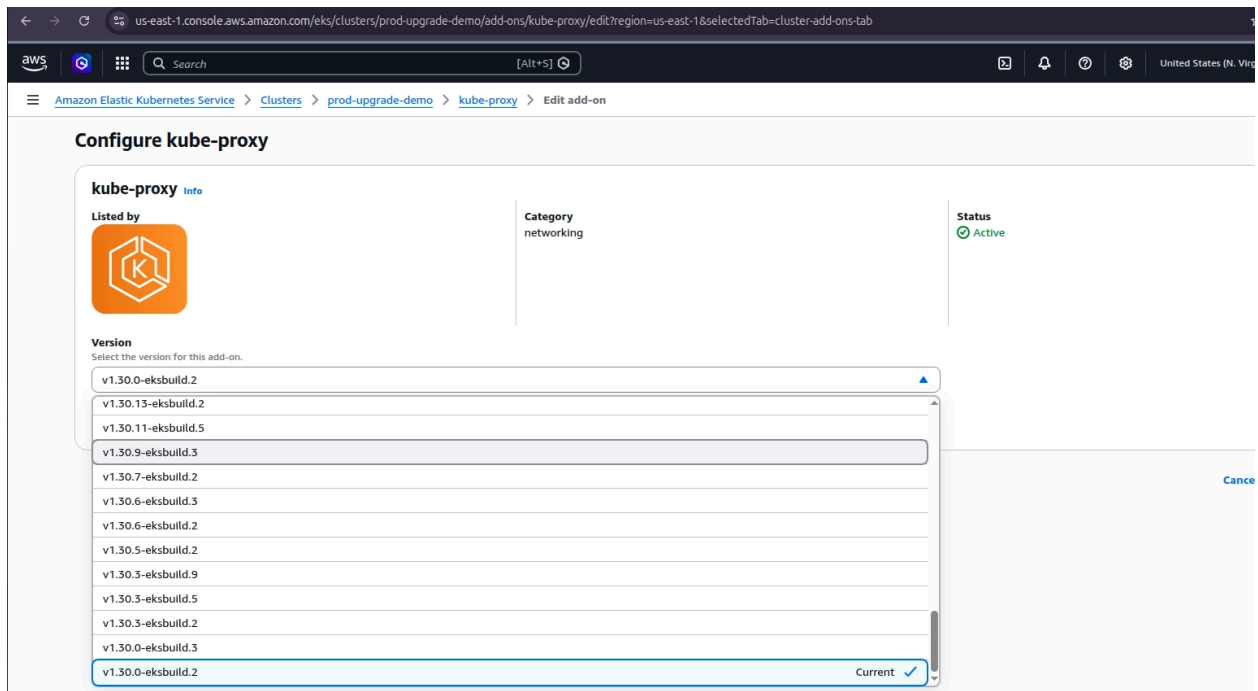
```
eksctl create cluster \  
  --name prod-upgrade-demo \  
  --region us-east-1 \  
  --version 1.33 \  
  --nodegroup-name ng-1 \  
  --node-type t3a.xlarge \  
  --nodes 2 \  
  --managed
```

```
eksctl create cluster --name prod-upgrade-demo --region ap-so  
uth-1 --version 1.33 --nodegroup-name ng-1 --node-type t3a.xl  
arge --nodes 2 --managed
```

Wait for the cluster to be available.

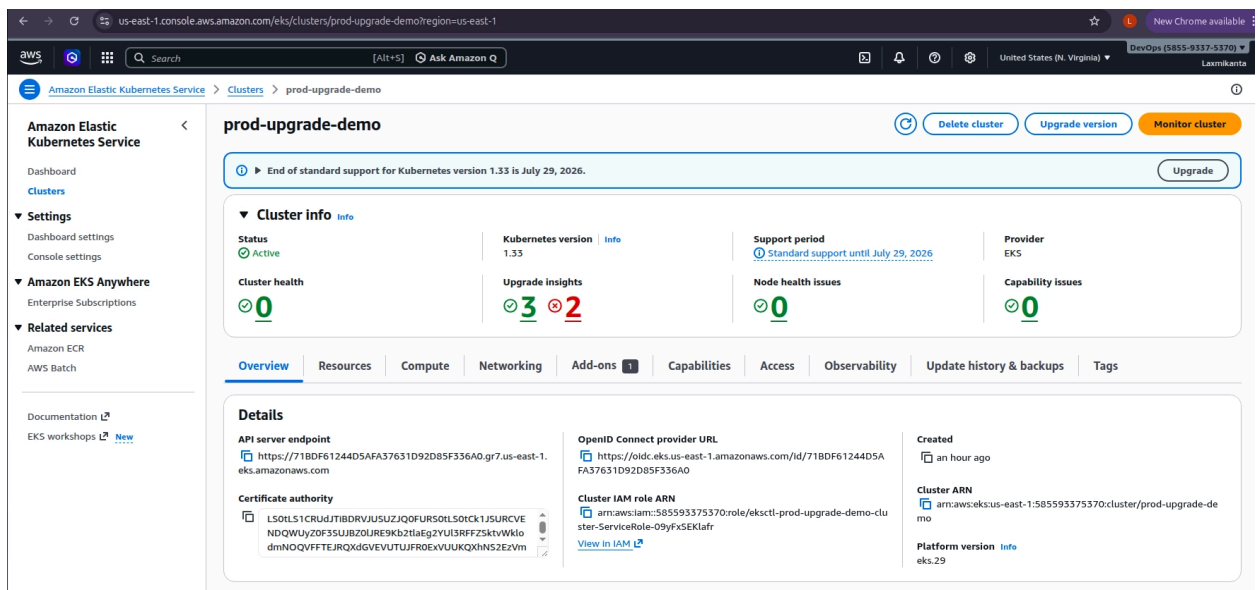
Then go tot the EKS UI → Addons → kube-proxy → Downgrade the the version choose the oldest one.

So that We will have some errors in the `upgrade insights` .

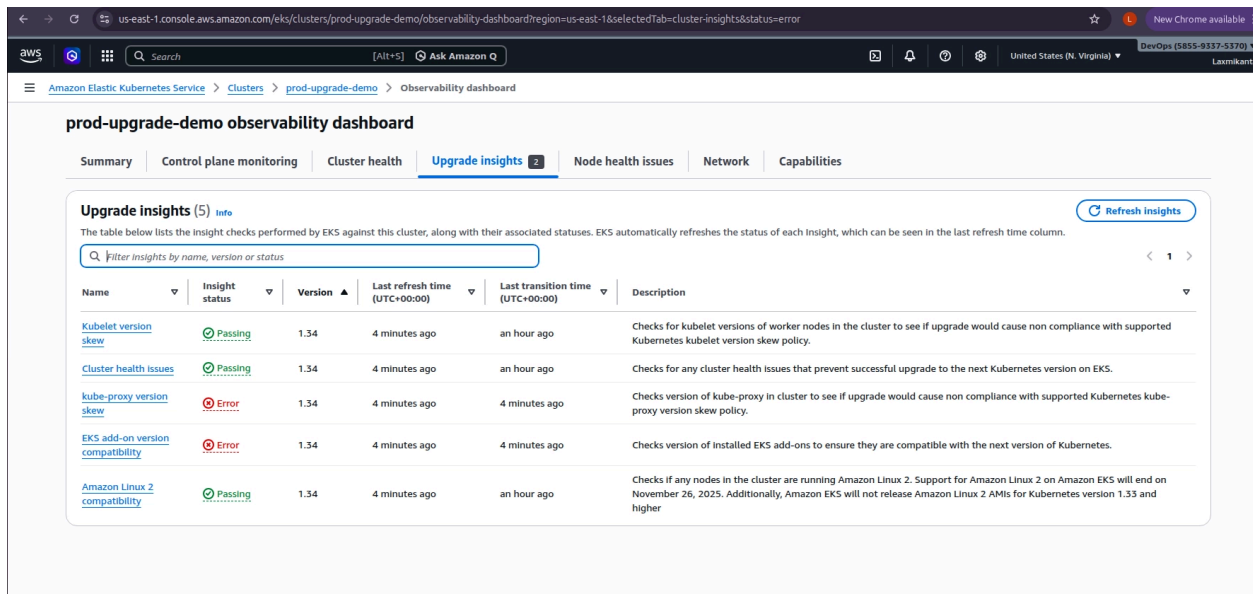


Now head to the upgrade insights in the EKS home page and click on the **refresh insight** button.

Wait for some time then it should 2 errors like below.



Click on the errors



Import the kube config to your bastion host or local machine.

```
aws eks update-kubeconfig --name prod-upgrade-demo --region us-east-1
```

Even i have the kubectl 1.33 version installed in my local machine

```
ubuntu@ubuntu-desktop: ~/MLOps-Essentials/9-kubeflow/7-project-kubflow... x ubuntu@ubuntu-desktop: ~/MLOps-Essentials/9-kubeflow/7-project-kubflow... x ubuntu@ubuntu-desktop: ~/Upgrade-EKS-cluster-production$ kubectl version
Client Version: v1.33.0
Kustomize Version: v5.6.0
Server Version: v1.33.8-eks-3a10415
ubuntu@ubuntu-desktop: ~/Upgrade-EKS-cluster-production$
```

Add some workloads

```
kubectl create deployment nginx --image=nginx
kubectl create deployment apache --image=apache
```

Now our scenarion is ready lets see the upgrade methods.

Lets say your manager tells you to upgrade the kubernetes cluster without having a downtime.

And currently you have all the worloads running in your cluster.

It's good to take the screenshots before upgrading the cluster. So that you will have the proof the state of the clusters.

Screenshot 1:

us-east-1.console.aws.amazon.com/eks/clusters/prod-upgrade-demo?region=us-east-1

Amazon Elastic Kubernetes Service > Clusters > prod-upgrade-demo

prod-upgrade-demo

End of standard support for Kubernetes version 1.33 is July 29, 2026.

Cluster info

Status: Active

Kubernetes version: 1.33

Support period: Standard support until July 29, 2026

Provider: EKS

Cluster health: 0

Upgrade insights: 3 2

Node health issues: 0

Capability issues: 0

Overview Resources Compute Networking Add-ons Capabilities Access Observability Update history & backups Tags

Details

API server endpoint: https://718DF61244D5AFA37631D92D85F336A0.gr7.us-east-1.eks.amazonaws.com

OpenID Connect provider URL: https://oidc.eks.us-east-1.amazonaws.com/id/718DF61244D5AFA37631D92D85F336A0

Certificate authority: LS0HLS1CRUdJTBDRVJUSUZQOFURSOLSOICK1JUSURVCE NDQWUyZ0F3SUJBZ0U9RE9Kb2tlaGZyUUI3RFFZSktvWklo dmhNQVFFTEJRQXGdGVVUTUJR0ExVUUKQXhNS2EzVm

Cluster IAM role ARN: arn:aws:iam::585593375370:role/eksctl-prod-upgrade-demo-cluster-ServiceRole-09yFxEKlafr

Created: an hour ago

Cluster ARN: arn:aws:eks:us-east-1:585593375370:cluster/prod-upgrade-demo

Platform version: v1.33

Screenshot 2:

us-east-1.console.aws.amazon.com/eks/clusters/prod-upgrade-demo/observability-dashboard?region=us-east-1&selectedTab=cluster-insights&status=error

Amazon Elastic Kubernetes Service > Clusters > prod-upgrade-demo > Observability dashboard

prod-upgrade-demo observability dashboard

Summary Control plane monitoring Cluster health Upgrade insights 2 Node health issues Network Capabilities

Upgrade insights (5)

The table below lists the insight checks performed by EKS against this cluster, along with their associated statuses. EKS automatically refreshes the status of each insight, which can be seen in the last refresh time column.

Filter insights by name, version or status

Name	Insight status	Version	Last refresh time (UTC+00:00)	Last transition time (UTC+00:00)	Description
Kubelet version skew	Passing	1.34	4 minutes ago	an hour ago	Checks for kubelet versions of worker nodes in the cluster to see if upgrade would cause non compliance with supported Kubernetes kubelet version skew policy.
Cluster health issues	Passing	1.34	4 minutes ago	an hour ago	Checks for any cluster health issues that prevent successful upgrade to the next Kubernetes version on EKS.
kube-proxy version skew	Error	1.34	4 minutes ago	4 minutes ago	Checks for kube-proxy version in cluster to see if upgrade would cause non compliance with supported Kubernetes kube-proxy version skew policy.
EKS add-on version compatibility	Error	1.34	4 minutes ago	4 minutes ago	Checks version of installed EKS add-ons to ensure they are compatible with the next version of Kubernetes.
Amazon Linux 2 compatibility	Passing	1.34	4 minutes ago	an hour ago	Checks if any nodes in the cluster are running Amazon Linux 2. Support for Amazon Linux 2 on Amazon EKS will end on November 26, 2025. Additionally, Amazon EKS will not release Amazon Linux 2 AMIs for Kubernetes version 1.33 and higher

Screenshot 3 (node group):

The screenshot shows the Amazon Elastic Kubernetes Service (EKS) console. The left sidebar contains navigation links for Dashboard, Clusters, Settings, Amazon EKS Anywhere, and Related services. The main content area is titled 'prod-upgrade-demo' and has tabs for Overview, Resources, Compute, Networking, Add-ons, Capabilities, Access, Observability, Update history & backups, and Tags. The 'Compute' tab is selected, showing 'Nodes (2)' and 'Node groups (1)'. The 'Nodes' section lists two nodes with names like 'ip-192-168-23-201.ec2.internal' and 'ip-192-168-33-51.ec2.internal', both in 'Ready' status. The 'Node groups' section shows a single node group named 'ng-1' with a desired size of 2 and an 'Active' status.

Screenshot 4 (with version):

The screenshot shows the Amazon Elastic Kubernetes Service (EKS) console. The left sidebar contains navigation links for Dashboard, Clusters, Settings, Amazon EKS Anywhere, and Related services. The main content area is titled 'prod-upgrade-demo' and has tabs for Overview, Resources, Compute, Networking, Add-ons, Capabilities, Access, Observability, Update history & backups, and Tags. The 'Add-ons' tab is selected, showing a list of add-ons. A notification at the top states 'New versions are available for 3 add-ons.' The add-ons listed are Metrics Server, kube-proxy, Amazon VPC CNI, and CoreDNS, all with 'Active' status. The 'Metrics Server' add-on is highlighted, showing its category as 'observability', status as 'Active', version as 'v0.8.1-eksbuild.2', and IAM role as 'Not required'.

Screenshot 5 (All pods state):

```
ubuntu@ubuntu-desktop: ~/MLops-Essentials/9-kubeflow/7-project-kubflow... x ubuntu@ubuntu-desktop: ~/MLops-Essentials/9-kubeflow/7-project-kubflow... x ubuntu@ubuntu-desktop: ~/Upgrade-EKS-cluster-production x
ubuntu@ubuntu-desktop: ~/Upgrade-EKS-cluster-production$ k get po -A
NAMESPACE   NAME                                     READY   STATUS    RESTARTS   AGE
default     apache-747cf8f6d7-j26l7               0/1     ImagePullBackOff    0          4m32s
default     nginx-5869d7778c-z88s2                1/1     Running      0          5m15s
kube-system aws-node-875sk                       2/2     Running      0          61m
kube-system aws-node-pgz6s                2/2     Running      0          61m
kube-system coredns-5d849c4789-k6sgw      1/1     Running      0          65m
kube-system coredns-5d849c4789-n8w5r      1/1     Running      0          65m
kube-system kube-proxy-9n25t             1/1     Running      0          20m
kube-system kube-proxy-xfw7q             1/1     Running      0          20m
kube-system metrics-server-7c9d75d7b-68lcf 1/1     Running      0          59m
kube-system metrics-server-7c9d75d7b-zw6v5 1/1     Running      0          59m
ubuntu@ubuntu-desktop: ~/Upgrade-EKS-cluster-production$
```

Screenshot 6 (All nodes state)

```
ubuntu@ubuntu-desktop: ~/MLops-Essentials/9-kubeflow/7-project-kubflow... x ubuntu@ubuntu-desktop: ~/MLops-Essentials/9-kubeflow/7-project-kubflow... x ubuntu@ubuntu-desktop: ~/Upgrade-EKS-cluster-production x
ubuntu@ubuntu-desktop: ~/Upgrade-EKS-cluster-production$ k get nodes
NAME                                     STATUS   ROLES    AGE   VERSION
ip-192-168-23-201.ec2.internal          Ready    <none>    62m   v1.33.8-eks-f69f56f
ip-192-168-33-51.ec2.internal           Ready    <none>    62m   v1.33.8-eks-f69f56f
ubuntu@ubuntu-desktop: ~/Upgrade-EKS-cluster-production$
ubuntu@ubuntu-desktop: ~/Upgrade-EKS-cluster-production$
```

Now once you have Proof we can proceed for upgradation.

The hirarchy would be

Control plane → Addons → Node group → Workloads (optional)

Before upgrading the control plane you must fix the **Upgrade Insight** Errors , if there is any.

Otherwise things might break.

As we are getting this kube-proxy version skew and incompatibility issue. We must fix it by upgrading the kube-proxy addons version.

prod-upgrade-demo observability dashboard

Summary | Control plane monitoring | Cluster health | **Upgrade insights** | Node health issues | Network | Capabilities

Upgrade insights (5) [Info](#) [Refresh insights](#)

The table below lists the insight checks performed by EKS against this cluster, along with their associated statuses. EKS automatically refreshes the status of each insight, which can be seen in the last refresh time column.

Filter insights by name, version or status 2 matches

Insight status = ERROR X Clear filters

Name	Insight status	Version	Last refresh time (UTC+00:00)	Last transition time (UTC+00:00)	Description
kube-proxy version skew	Error	1.34	27 minutes ago	27 minutes ago	Checks version of kube-proxy in cluster to see if upgrade would cause non compliance with supported Kubernetes kube-proxy version skew policy.
EKS add-on version compatibility	Error	1.34	27 minutes ago	27 minutes ago	Checks version of installed EKS add-ons to ensure they are compatible with the next version of Kubernetes.

Take the latest one for the current version and then update.

Configure kube-proxy

kube-proxy [Info](#)

Listed by 

Category networking

Status Active

Version
Select the version for this add-on.

v1.33.8-eksbuild.4 Current/Latest ✓

v1.33.7-eksbuild.2

v1.33.5-eksbuild.2

v1.33.3-eksbuild.10

v1.33.3-eksbuild.6

v1.33.3-eksbuild.4

v1.33.0-eksbuild.2

v1.32.11-eksbuild.5

v1.32.11-eksbuild.2

v1.32.9-eksbuild.2

v1.32.6-eksbuild.12

Default

Cancel Save changes

At the same time check if new pod comes in the cluster.

```
ubuntu@ubuntu-desktop: ~/MLOps-Essentials/9-kubeflow/7-project-kubeflow... x ubuntu@ubuntu-desktop: ~/MLOps-Essentials/9-kubeflow/7-project-kubeflow... x ubuntu@ubuntu-desktop: ~/Upgrade-EKS-cluster-production x
ubuntu@ubuntu-desktop: ~/Upgrade-EKS-cluster-production$ k get nodes
NAME                                STATUS    ROLES    AGE   VERSION
ip-192-168-23-201.ec2.internal      Ready    <none>    62m   v1.33.8-eks-f69f56f
ip-192-168-33-51.ec2.internal       Ready    <none>    62m   v1.33.8-eks-f69f56f
ubuntu@ubuntu-desktop: ~/Upgrade-EKS-cluster-production$ k get po -n kube-system | grep proxy
kube-proxy-cftlb                    1/1      Running   0      10s
kube-proxy-v6gcz                    1/1      Running   0      14s
ubuntu@ubuntu-desktop: ~/Upgrade-EKS-cluster-production$
```

Check the logs if any error or failing happens.

Make sure no error happens.

```
ubuntu@ubuntu-desktop: ~/MLOps-Essentials/9-kubeflow/7-project-kubeflow... x ubuntu@ubuntu-desktop: ~/MLOps-Essentials/9-kubeflow/7-project-kubeflow... x ubuntu@ubuntu-desktop: ~/Upgrade-EKS-cluster-production x
set nodePortAddresses (-nodeport-addresses) to filter loopback addresses" ipFamily="IPv4"
I0311 12:43:07.661422 1 proxier.go:266] "Using iptables mark for masquerade" ipFamily="IPv4" mark="0x00004000"
I0311 12:43:07.661657 1 proxier.go:311] "Iptables sync params" ipFamily="IPv4" minSyncPeriod="1s" syncPeriod="30s" burstSyncs=2
I0311 12:43:07.661725 1 proxier.go:321] "Iptables supports --random-fully" ipFamily="IPv4"
I0311 12:43:07.661889 1 proxier.go:266] "Using iptables mark for masquerade" ipFamily="IPv6" mark="0x00004000"
I0311 12:43:07.661980 1 proxier.go:311] "Iptables sync params" ipFamily="IPv6" minSyncPeriod="1s" syncPeriod="30s" burstSyncs=2
I0311 12:43:07.662039 1 proxier.go:321] "Iptables supports --random-fully" ipFamily="IPv6"
I0311 12:43:07.662087 1 server.go:516] "Version info" version="v1.33.8-2+97c2604d2fcd20"
I0311 12:43:07.662173 1 server.go:518] "Golang settings" GOGC="" GOMAXPROCS="" GOTRACEBACK=""
I0311 12:43:07.664799 1 config.go:105] "Starting endpoint slice config controller"
I0311 12:43:07.664860 1 shared_informer.go:350] "Waiting for caches to sync" controller="endpoint slice config"
I0311 12:43:07.664862 1 config.go:440] "Starting serviceCIDR config controller"
I0311 12:43:07.664897 1 shared_informer.go:350] "Waiting for caches to sync" controller="serviceCIDR config"
I0311 12:43:07.665102 1 config.go:199] "Starting service config controller"
I0311 12:43:07.665119 1 shared_informer.go:350] "Waiting for caches to sync" controller="service config"
I0311 12:43:07.665793 1 config.go:329] "Starting node config controller"
I0311 12:43:07.665817 1 shared_informer.go:350] "Waiting for caches to sync" controller="node config"
I0311 12:43:07.669329 1 reflector.go:430] "Caches populated" types="v1.Service" reflector="k8s.io/client-go/informers/factory.go:160"
I0311 12:43:07.670092 1 reflector.go:430] "Caches populated" types="v1.EndpointSlice" reflector="k8s.io/client-go/informers/factory.go:160"
I0311 12:43:07.669373 1 reflector.go:430] "Caches populated" types="v1.Node" reflector="k8s.io/client-go/informers/factory.go:160"
I0311 12:43:07.670661 1 reflector.go:430] "Caches populated" types="v1.ServiceCIDR" reflector="k8s.io/client-go/informers/factory.go:160"
I0311 12:43:07.669502 1 proxier.go:793] "Not syncing iptables until Services and Endpoints have been received from master" ipFamily="IPv6"
I0311 12:43:07.671704 1 proxier.go:793] "Not syncing iptables until Services and Endpoints have been received from master" ipFamily="IPv4"
I0311 12:43:07.765583 1 shared_informer.go:357] "Caches are synced" controller="endpoint slice config"
I0311 12:43:07.765619 1 shared_informer.go:357] "Caches are synced" controller="serviceCIDR config"
I0311 12:43:07.765645 1 proxier.go:793] "Not syncing iptables until Services and Endpoints have been received from master" ipFamily="IPv4"
I0311 12:43:07.765668 1 proxier.go:793] "Not syncing iptables until Services and Endpoints have been received from master" ipFamily="IPv6"
I0311 12:43:07.765552 1 shared_informer.go:357] "Caches are synced" controller="service config"
I0311 12:43:07.765801 1 proxier.go:815] "Syncing iptables rules" ipFamily="IPv4" fullSync=true
I0311 12:43:07.766053 1 shared_informer.go:357] "Caches are synced" controller="node config"
I0311 12:43:07.870529 1 proxier.go:1536] "Reloading service iptables data" ipFamily="IPv4" numServices=6 numEndpoints=11 numFilterChains=6 numFilterRules=4 numNATChains=4
numNATRules=44
I0311 12:43:08.003447 1 proxier.go:809] "SyncProxyRules complete" ipFamily="IPv4" elapsed="237.750899ms"
I0311 12:43:08.003518 1 proxier.go:815] "Syncing iptables rules" ipFamily="IPv6" fullSync=true
I0311 12:43:08.133064 1 proxier.go:1536] "Reloading service iptables data" ipFamily="IPv6" numServices=0 numEndpoints=0 numFilterChains=5 numFilterRules=3 numNATChains=4
umNATRules=5
I0311 12:43:08.250009 1 proxier.go:809] "SyncProxyRules complete" ipFamily="IPv6" elapsed="246.488402ms"
```

If there would be any error then revert back and fix the error or find cause of error and then update it again. As it's production you should be very confident what you are doing.

Similarly there can be other issues in the **upgrade insights** Fix all of them make sure the errors are gone then you can upgrade the **control plane**.

Now head EKS home UI click on the Upgrade Insights and **refresh**.

The screenshot shows the AWS EKS console's 'prod-upgrade-demo observability dashboard'. A blue banner at the top indicates 'Add-on update in progress'. The 'Upgrade insights' tab is selected, showing a table with two error insights. A 'Refresh insights' button is in the top right.

Name	Insight status	Version	Last refresh time (UTC+00:00)	Last transition time (UTC+00:00)	Description
kube-proxy version skew	Error	1.34	33 minutes ago	33 minutes ago	Checks version of kube-proxy in cluster to see if upgrade would cause non compliance with supported Kubernetes kube-proxy version skew policy.
EKS add-on version compatibility	Error	1.34	33 minutes ago	33 minutes ago	Checks version of installed EKS add-ons to ensure they are compatible with the next version of Kubernetes.

Wait for some time.

Make sure its all PASSING

The screenshot shows the same AWS EKS console dashboard after a refresh. All insights are now in a 'Passing' state, indicated by green checkmarks. A 'Refresh insights' button is still present.

Name	Insight status	Version	Last refresh time (UTC+05:30)	Last transition time (UTC+05:30)	Description
Amazon Linux 2 compatibility	Passing	1.34	5 hours ago	6 hours ago	Checks if any nodes in the cluster are running Amazon Linux 2. Support for Amazon Linux 2 on Amazon EKS will end on November 26, 2025. Additionally, Amazon EKS will not release Amazon Linux 2 AMIs for Kubernetes version 1.33 and higher
Cluster health issues	Passing	1.34	5 hours ago	6 hours ago	Checks for any cluster health issues that prevent successful upgrade to the next Kubernetes version on EKS.
EKS add-on version compatibility	Passing	1.34	5 hours ago	5 hours ago	Checks version of installed EKS add-ons to ensure they are compatible with the next version of Kubernetes.
Kubelet version skew	Passing	1.34	5 hours ago	6 hours ago	Checks for kubelet versions of worker nodes in the cluster to see if upgrade would cause non compliance with supported Kubernetes kubelet version skew policy.

Amazon Elastic Kubernetes Service > Clusters > prod-upgrade-demo

prod-upgrade-demo Delete cluster Upgrade version Monitor cluster

End of standard support for Kubernetes version 1.33 is July 29, 2026. Upgrade

Cluster info			
Status Active	Kubernetes version 1.33	Support period Standard support until July 29, 2026	Provider EKS
Cluster health 0	Upgrade insights 5	Node health issues 0	Capability issues 0

Details

API server endpoint https://718DF61244D5AFA37631D92D85F336A0.gr7.us-east-1.eks.amazonaws.com	OpenID Connect provider URL https://oidc.eks.us-east-1.amazonaws.com/id/718DF61244D5AFA37631D92D85F336A0	Created 6 hours ago
--	--	------------------------

Lets Upgrade the cluster now by clicking on the upgrade button in the UI

Amazon Elastic Kubernetes Service > Clusters > prod-upgrade-demo

prod-upgrade-demo Delete cluster Upgrade version Monitor cluster

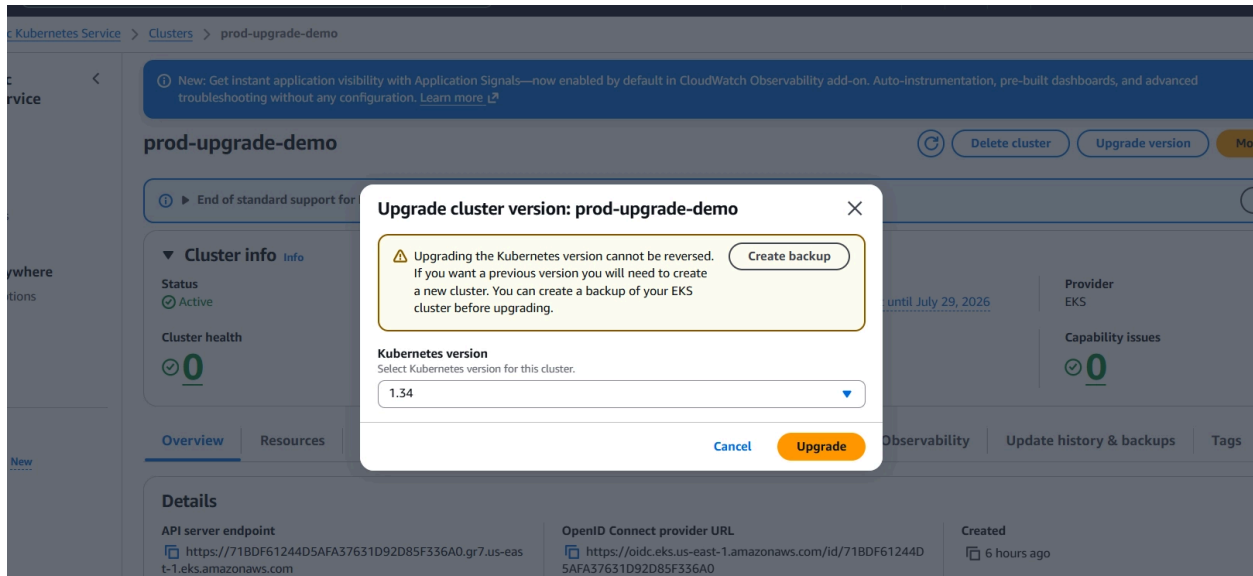
End of standard support for Kubernetes version 1.33 is July 29, 2026. Upgrade

Cluster info			
Status Active	Kubernetes version 1.33	Support period Standard support until July 29, 2026	Provider EKS
Cluster health 0	Upgrade insights 5	Node health issues 0	Capability issues 0

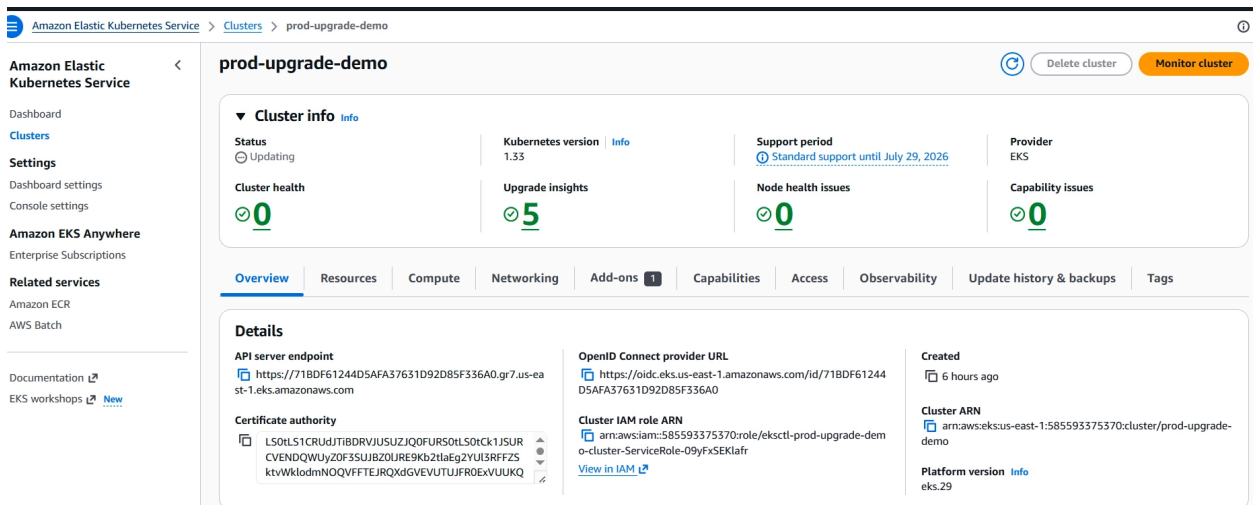
Details

API server endpoint https://718DF61244D5AFA37631D92D85F336A0.gr7.us-east-1.eks.amazonaws.com	OpenID Connect provider URL https://oidc.eks.us-east-1.amazonaws.com/id/718DF61244D5AFA37631D92D85F336A0	Created 6 hours ago
--	--	------------------------

Choose Version



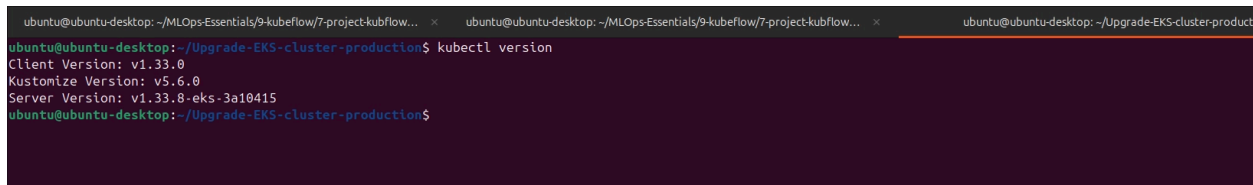
Now its being updated



It takes some time to upgrade , meanwhile you can update the kubectl

Currently we have 1.33

We will upgrade to 1.34



Update using below commands

```
sudo rm /etc/apt/sources.list.d/google-cloud-sdk.list
echo 'deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg] https://pkgs.k8s.io/core:/stable:/v1.34/deb/ /' | sudo tee /etc/apt/sources.list.d/kubernetes.list
sudo chmod 644 /etc/apt/sources.list.d/kubernetes.list
curl -fsSL https://pkgs.k8s.io/core:/stable:/v1.34/deb/Release.key | sudo gpg --dearmor -o /etc/apt/keyrings/kubernetes-apt-keyring.gpg
sudo chmod 644 /etc/apt/keyrings/kubernetes-apt-keyring.gpg
```

```
sudo apt update
```

Now check available kubectl version:

```
apt-cache madison kubectl
```

Expected Output:

```
kubectl | 1.34.5-1.1 | https://pkgs.k8s.io/core:/stable:/v1.34/deb Packages
kubectl | 1.34.4-1.1 | https://pkgs.k8s.io/core:/stable:/v1.34/deb Packages
kubectl | 1.34.3-1.1 | https://pkgs.k8s.io/core:/stable:/v1.34/deb Packages
kubectl | 1.34.2-1.1 | https://pkgs.k8s.io/core:/stable:/v1.34/deb Packages
kubectl | 1.34.1-1.1 | https://pkgs.k8s.io/core:/stable:/v1.34/deb Packages
kubectl | 1.34.0-1.1 | https://pkgs.k8s.io/core:/stable:/v1.34/deb Packages
```

Choose the most update one.


```
sudo apt install kubect=1.34.5-1.1
```

Or simple you can run the below command it takes the updated one.

```
sudo apt install kubect
```

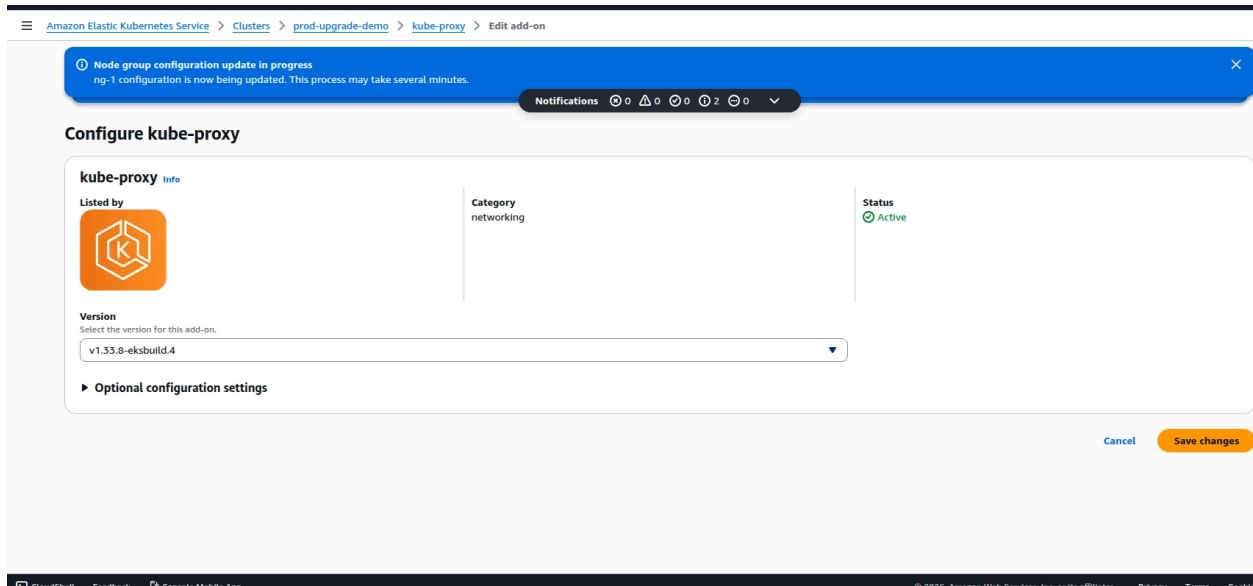
```
get:1 https://prod-con.packages.k8s.io/repositories/tsv:/kubernetes:/core:/stable:/v1.34/deb- Kubect=1.34.5-1.1 [11.7 MB]
Fetched 11.7 MB in 6s (1,849 kB/s)
(Reading database ... 268330 files and directories currently installed.)
Preparing to unpack .../kubect=1.34.5-1.1_amd64.deb ...
Unpacking kubect (1.34.5-1.1) over (1.33.0-1.1) ...
Setting up kubect (1.34.5-1.1) ...
ubuntu@ubuntu-desktop:~/Upgrade-EKS-cluster-production$ kubect version
Client Version: v1.34.5
Kustomize Version: v5.7.1
Server Version: v1.34.4-eks-3a10415
ubuntu@ubuntu-desktop:~/Upgrade-EKS-cluster-production$
```

The control plane has upgraded to 1.34 you can see in the UI

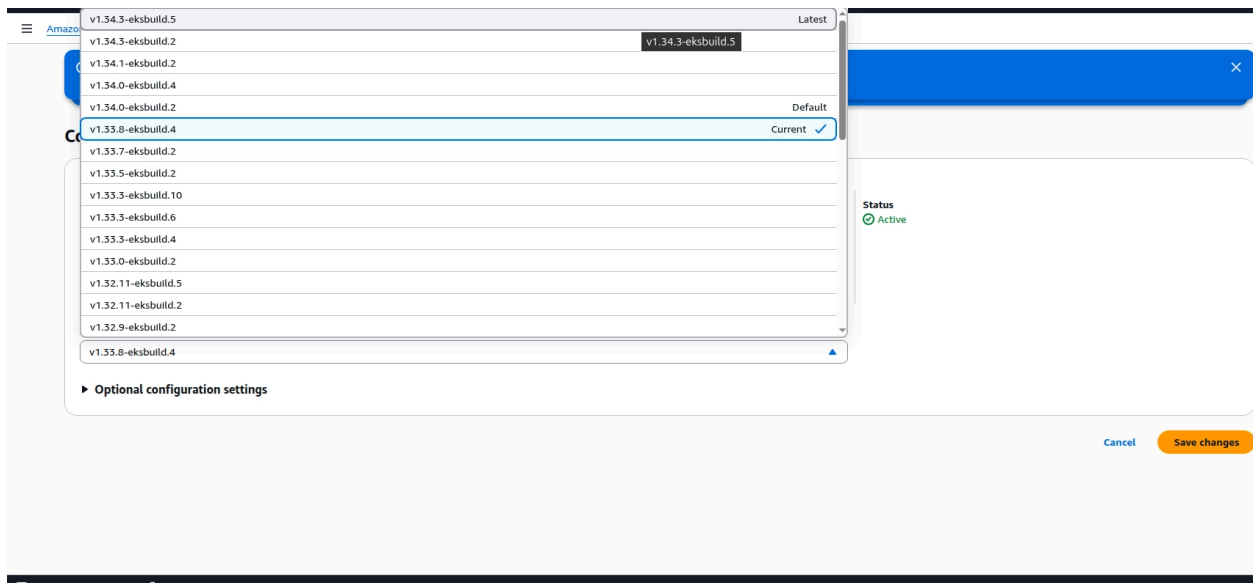
The screenshot shows the Amazon EKS console for the 'prod-upgrade-demo' cluster. The left sidebar lists navigation options like 'Amazon Elastic Kubernetes Service', 'Clusters', 'prod-upgrade-demo', 'hboard', 'stors', 'tings', 'hboard settings', 'hboard settings', 'hboard settings', 'Amazon EKS Anywhere', 'erprise Subscriptions', 'ated services', 'Amazon ECR', 'S Batch', 'umentation', and 'workshops'. The main content area shows the cluster's status as 'Active' and 'Kubernetes version 1.34'. It also displays 'Cluster health' with 0 issues, 'Upgrade insights' with 5 insights, 'Node health issues' with 0 issues, and 'Capability issues' with 0 issues. The 'Details' section includes the API server endpoint, OpenID Connect provider URL, Certificate authority, Cluster IAM role ARN, and Created time (7 hours ago). The 'Platform version' is also shown as 'v1.34.5-1.1'.

Lets Upgrade the addons one by one.

Kube-proxy



Choose the latest one.



Hit save changes.

At the same time check in your cluster if new pods comes in.

```
kubectl -n kube-system get po -w | grep proxy
```

Expected Outputs:

kube-proxy-nzqtb	1/1	Running	0
11s			
kube-proxy-wdxgl	1/1	Running	0
8s			

Then check logs, make sure there no recent error logs. Make sure to check the logging time.

```
kubectl -n kube-system logs kube-proxy-nzqtb -f
```

Expected:

```
.....
I0312 04:50:20.922165      1 proxier.go:763] "Syncing iptabl
es rules" ipFamily="IPv4" fullSync=true
I0312 04:50:20.922408      1 shared_informer.go:356] "Caches
are synced" controller="node config"
I0312 04:50:21.017984      1 proxier.go:1484] "Reloading ser
vice iptables data" ipFamily="IPv4" numServices=6 numEndpoint
s=11 numFilterChains=6 numFilterRules=4 numNATChains=21 numNA
TRules=44
I0312 04:50:21.151017      1 proxier.go:757] "SyncProxyRules
complete" ipFamily="IPv4" elapsed="229.014377ms"
I0312 04:50:21.151067      1 proxier.go:763] "Syncing iptabl
es rules" ipFamily="IPv6" fullSync=true
I0312 04:50:21.269240      1 proxier.go:1484] "Reloading ser
vice iptables data" ipFamily="IPv6" numServices=0 numEndpoint
s=0 numFilterChains=5 numFilterRules=3 numNATChains=4 numNATR
ules=5
I0312 04:50:21.391836      1 proxier.go:757] "SyncProxyRules
complete" ipFamily="IPv6" elapsed="240.760951ms"
```

Always compare logging time with current time.

Go back to the UI refresh, it should be successful..

The screenshot shows the Amazon Elastic Kubernetes Service console. The breadcrumb navigation at the top reads: Amazon Elastic Kubernetes Service > Clusters > prod-upgrade-demo > kube-proxy > e554ec69-b330-3269-a3e4-93b0e376bada. The left sidebar contains the following menu items: Amazon Elastic Kubernetes Service, Dashboard, Clusters, Settings (with sub-items: Dashboard settings, Console settings), Amazon EKS Anywhere (with sub-item: Enterprise Subscriptions), Related services (with sub-items: Amazon ECR, AWS Batch), Documentation, and EKS workshops. The main content area features a blue notification banner at the top stating 'Add-on update in progress' and 'The kube-proxy add-on is now being updated. This process may take several minutes.' Below this, the 'Update ID: e554ec69-b330-3269-a3e4-93b0e376bada' is displayed. The 'General configuration' section shows: Update ID (e554ec69-b330-3269-a3e4-93b0e376bada), Status (successful), Type (AddonUpdate), AddonVersion (v1.34.3-eksbuild.5), ResolveConflicts (NONE), and PodIdentityAssociations (empty array). The 'Errors (0)' section shows a table with columns 'Error code', 'Error message', and 'Resource IDs', containing the message 'No errors. There are no errors for this update.'

Follow the same process for each addons update.

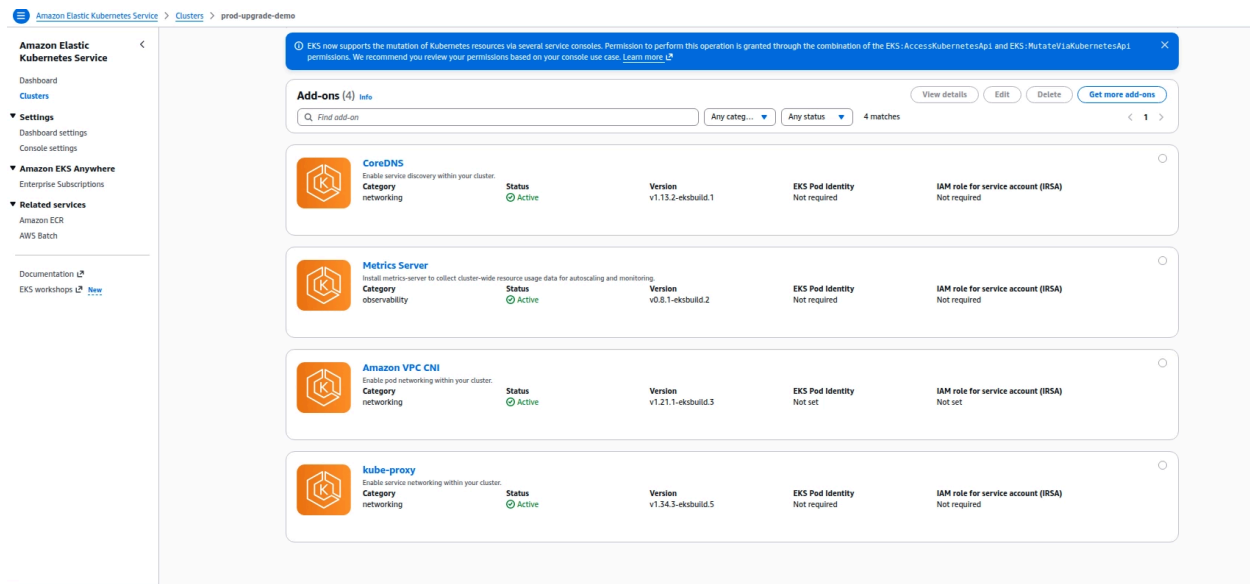
Update the addons by following this above step, if something goes wrong then revert back fix the issue and repeat.



Ignore the addons which are already updated and the newest.

I have updated all the addons.

Make sure non of them shows update available.



Now its time to update the node group.

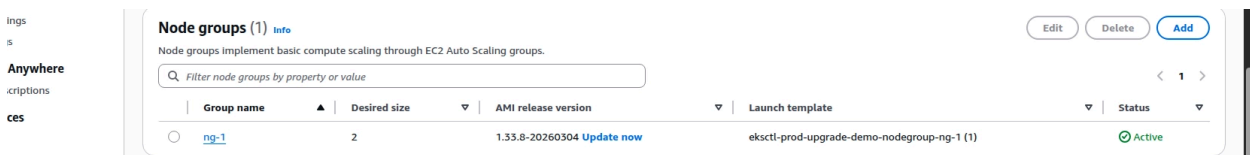
Our node group is still in v1.33

```
ubuntu@ubuntu-desktop:~/Upgrade-EKS-cluster-production$ kubectl get nodes
NAME                                STATUS    ROLES    AGE    VERSION
ip-192-168-23-215.ec2.internal      Ready    <none>   11m    v1.33.8-eks-f69f56f
ip-192-168-32-107.ec2.internal      Ready    <none>   10m    v1.33.8-eks-f69f56f
ubuntu@ubuntu-desktop:~/Upgrade-EKS-cluster-production$
```

Go to the Compute section in the EKS UI.

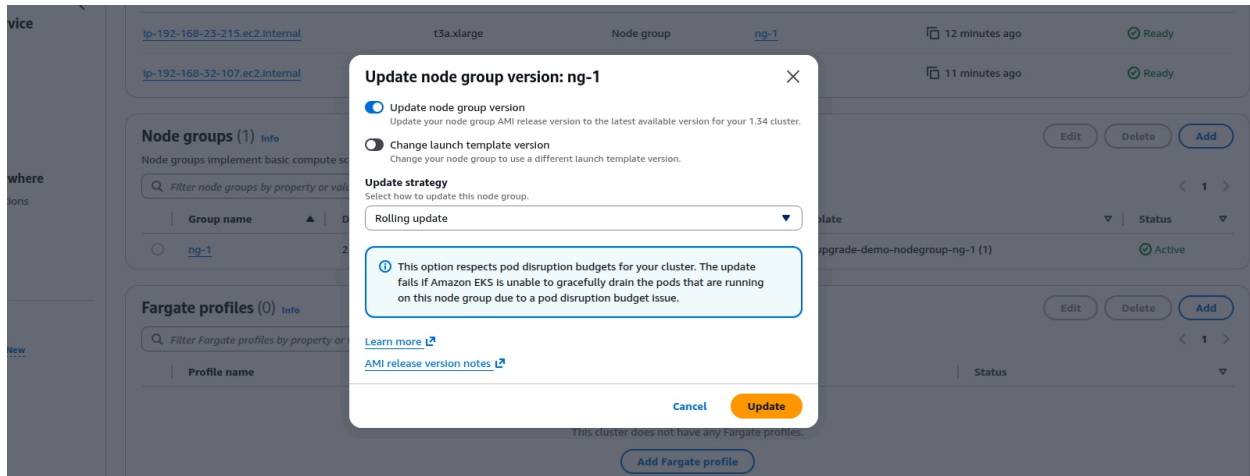
Scroll down to the node groups.

Click on update now button



Choose Rolling Update so that the new nodes comes first and the delete the older node, So we will get almost zero downtime.

If you have mutliple nodegroups then update one by one.



Then Update.

Now in the cluster check nodes activity.

It takes some time to update.

```
kubectl get nodes -w
```

Expected:

```
ubuntu@ubuntu-desktop:~/Upgrade-EKS-cluster-production$ kubectl get nodes -w
NAME                                STATUS    ROLES    AGE     VERSION
ip-192-168-23-215.ec2.internal      Ready    <none>   13m     v1.33.8-eks-f69f56f
ip-192-168-32-107.ec2.internal      Ready    <none>   13m     v1.33.8-eks-f69f56f
ip-192-168-23-215.ec2.internal      Ready    <none>   14m     v1.33.8-eks-f69f56f
ip-192-168-32-107.ec2.internal      Ready    <none>   14m     v1.33.8-eks-f69f56f
ip-192-168-33-16.ec2.internal        NotReady <none>   0s      v1.34.4-eks-f69f56f
ip-192-168-33-16.ec2.internal        NotReady <none>   0s      v1.34.4-eks-f69f56f
ip-192-168-33-16.ec2.internal        NotReady <none>   0s      v1.34.4-eks-f69f56f
ip-192-168-33-16.ec2.internal        NotReady <none>   0s      v1.34.4-eks-f69f56f
ip-192-168-33-16.ec2.internal        NotReady <none>   0s      v1.34.4-eks-f69f56f
ip-192-168-33-16.ec2.internal        NotReady <none>   0s      v1.34.4-eks-f69f56f
ip-192-168-26-223.ec2.internal        NotReady <none>   0s      v1.34.4-eks-f69f56f
ip-192-168-26-223.ec2.internal        NotReady <none>   0s      v1.34.4-eks-f69f56f
ip-192-168-26-223.ec2.internal        NotReady <none>   0s      v1.34.4-eks-f69f56f
ip-192-168-26-223.ec2.internal        NotReady <none>   0s      v1.34.4-eks-f69f56f
ip-192-168-26-223.ec2.internal        NotReady <none>   0s      v1.34.4-eks-f69f56f
ip-192-168-26-223.ec2.internal        NotReady <none>   0s      v1.34.4-eks-f69f56f
ip-192-168-26-223.ec2.internal        NotReady <none>   2s      v1.34.4-eks-f69f56f
ip-192-168-33-16.ec2.internal        NotReady <none>   3s      v1.34.4-eks-f69f56f
ip-192-168-33-16.ec2.internal        NotReady <none>   10s     v1.34.4-eks-f69f56f
ip-192-168-26-223.ec2.internal        NotReady <none>   10s     v1.34.4-eks-f69f56f
ip-192-168-33-16.ec2.internal        Ready    <none>   14s     v1.34.4-eks-f69f56f
ip-192-168-33-16.ec2.internal        Ready    <none>   14s     v1.34.4-eks-f69f56f
ip-192-168-26-223.ec2.internal        Ready    <none>   13s     v1.34.4-eks-f69f56f
ip-192-168-26-223.ec2.internal        Ready    <none>   13s     v1.34.4-eks-f69f56f
ip-192-168-33-16.ec2.internal        Ready    <none>   15s     v1.34.4-eks-f69f56f
ip-192-168-26-223.ec2.internal        Ready    <none>   14s     v1.34.4-eks-f69f56f
ip-192-168-26-223.ec2.internal        Ready    <none>   17s     v1.34.4-eks-f69f56f
ip-192-168-33-16.ec2.internal        Ready    <none>   18s     v1.34.4-eks-f69f56f
ip-192-168-33-16.ec2.internal        Ready    <none>   31s     v1.34.4-eks-f69f56f
ip-192-168-26-223.ec2.internal        Ready    <none>   30s     v1.34.4-eks-f69f56f
ip-192-168-32-107.ec2.internal        Ready,SchedulingDisabled <none>   15m     v1.33.8-eks-f69f56f
ip-192-168-32-107.ec2.internal        Ready,SchedulingDisabled <none>   15m     v1.33.8-eks-f69f56f
ip-192-168-23-215.ec2.internal        Ready,SchedulingDisabled <none>   15m     v1.33.8-eks-f69f56f
ip-192-168-23-215.ec2.internal        Ready,SchedulingDisabled <none>   15m     v1.33.8-eks-f69f56f
ip-192-168-32-107.ec2.internal        Ready,SchedulingDisabled <none>   15m     v1.33.8-eks-f69f56f
ip-192-168-26-223.ec2.internal        Ready    <none>   2m33s   v1.34.4-eks-f69f56f
```

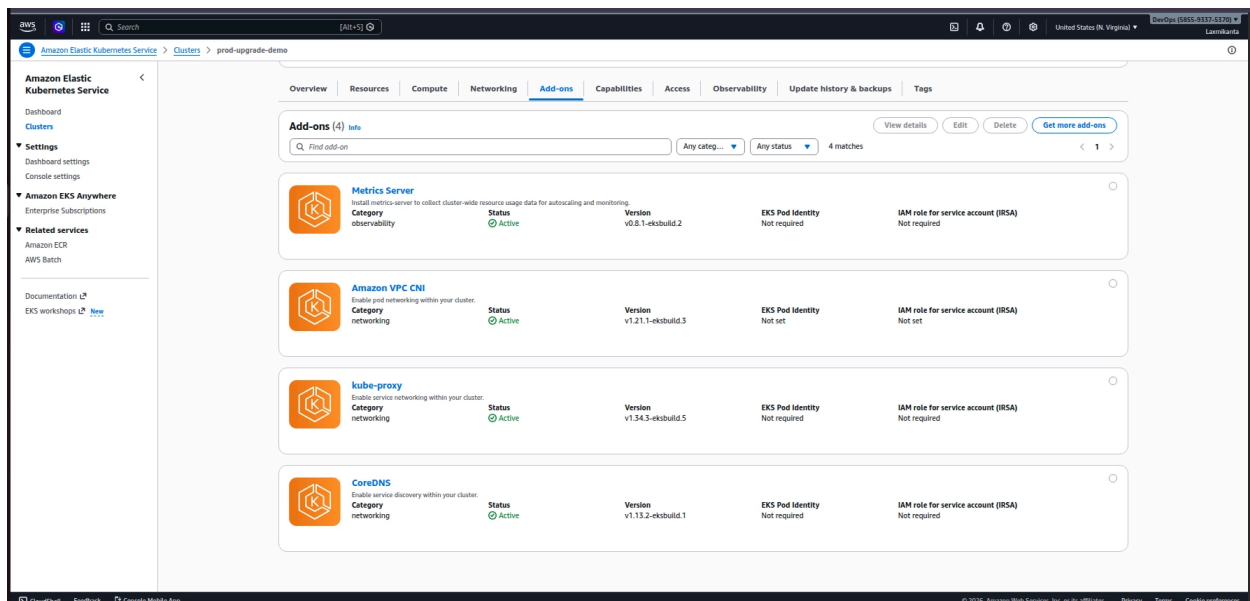
If you notice , First 2 new nodes joined the cluster and then worload shifted to those new nodes, then old version nodes are drained and went to not-ready state, At last they left the cluster and got deleted.

After your work is done make sure you take the screenshots and update in the ticket or keep it with yourself for proof.

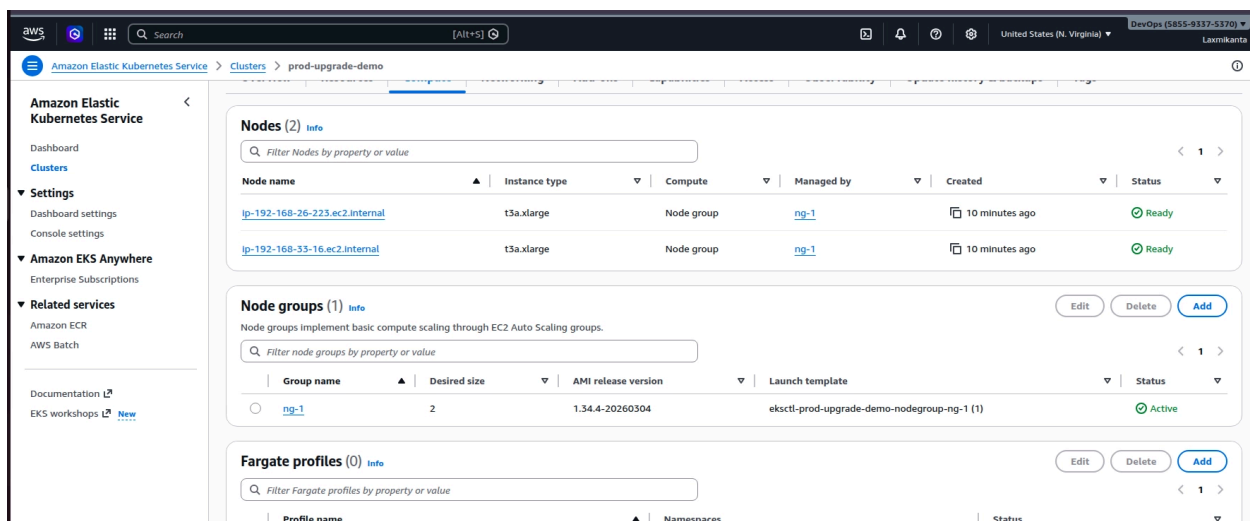
Cluster State:

The screenshot displays the Amazon Elastic Kubernetes Service (EKS) console for the 'prod-upgrade-demo' cluster. The interface includes a left-hand navigation menu with options like Dashboard, Clusters, Settings, Amazon EKS Anywhere, and Related services. The main content area shows the cluster's status as 'Active' and provides a summary of key metrics: Cluster health (0 issues), Upgrade insights (5), Node health issues (0), and Capability issues (0). A warning banner at the top indicates the end of standard support for Kubernetes version 1.34 on December 2, 2026. Below the summary, the 'Details' tab is selected, showing the API server endpoint, Certificate authority, OpenID Connect provider URL, Cluster IAM role ARN, and other configuration details. The cluster is running on the 'eks.15' platform version.

Addons:



Node groups and nodes:



```
ubuntu@ubuntu-desktop:~/Upgrade-EKS-cluster-production$ kubectl get nodes
NAME                                STATUS    ROLES    AGE    VERSION
ip-192-168-26-223.ec2.internal      Ready    <none>   9m13s  v1.34.4-eks-f69f56f
ip-192-168-33-16.ec2.internal      Ready    <none>   9m14s  v1.34.4-eks-f69f56f
ubuntu@ubuntu-desktop:~/Upgrade-EKS-cluster-production$
```

At last very important, make sure all the pods are running or the state in which they were before you upgrade the cluster.

`kubectl get pods -A`


```

ubuntu@ubuntu-desktop:~/Upgrade-EKS-cluster-production$ kubectl get nodes
NAME                                STATUS    ROLES    AGE    VERSION
ip-192-168-26-223.ec2.internal      Ready    <none>    9m13s  v1.34.4-eks-f69f56f
ip-192-168-33-16.ec2.internal      Ready    <none>    9m14s  v1.34.4-eks-f69f56f
ubuntu@ubuntu-desktop:~/Upgrade-EKS-cluster-production$ kubectl get po -A
NAMESPACE   NAME                                READY    STATUS    RESTARTS   AGE
default     apache-747cf8f6d7-hzd49            0/1      ImagePullBackOff    0           6m1s
default     nginx-5869d7778c-59n7t             1/1      Running         0           6m1s
kube-system aws-node-22nwn                2/2      Running         0           12m
kube-system aws-node-q7fwd            2/2      Running         0           12m
kube-system coredns-6fd7c76c7f-qwq8n    1/1      Running         0           10m
kube-system coredns-6fd7c76c7f-zk7w7    1/1      Running         0           9m3s
kube-system kube-proxy-2lzrp           1/1      Running         0           12m
kube-system kube-proxy-jqv6v           1/1      Running         0           12m
kube-system metrics-server-7c9d75d7b-27rw2 1/1      Running         0           5m
kube-system metrics-server-7c9d75d7b-xlh7k 1/1      Running         0           6m1s
ubuntu@ubuntu-desktop:~/Upgrade-EKS-cluster-production$

```

Kubectl version:

```

ubuntu@ubuntu-desktop:~/Upgrade-EKS-cluster-production$ kubectl version
Client Version: v1.34.5
Kustomize Version: v5.7.1
Server Version: v1.34.4-eks-3a10415
ubuntu@ubuntu-desktop:~/Upgrade-EKS-cluster-production$

```

You are done with the upgrade.